

Selection Discipline for Data-Dependent Interface Projections: Witness-Indexed Maps and Pilot/Frozen Phases for Receipt-Grade Audits

February 28, 2026 (archive v147)

Abstract

Several “interface projections” in this archive require choosing a discrete map from data: timing bucket boundaries (Synthesis 27), coarsening/partition maps for large alphabets (Synthesis 28), or “top- m plus other” hash bins for structured witnesses. If the chosen map is published, the selection itself can become an additional, reviewable exposure surface. This note centralizes two conservative patterns that keep papers non-redundant and receipts referee-grade: (i) *witness-indexed families* where the chosen map is a deterministic function of a *published witness* W already in scope, and (ii) *pilot/frozen phases* where map selection and audit evidence are separated by change control and spending (Synthesis 26).

1 Setting and scope

A deployment publishes a receipt claiming that an observed projection Y is nearly secret-independent (Anonymity B), and supports that claim with log-audit evidence (Synthesis 25) under a change-control discipline (Synthesis 26). [?, ?, ?] When the projection is derived as $Y = f_\theta(T)$ with a data-chosen parameter θ (bucket boundaries, partition rules), this note asks one narrow question: *how can θ be chosen and published without silently expanding the exposure surface?*

We keep the answer intentionally conservative. If a deployment wants to use sophisticated selection (e.g., adaptive data-driven partitions), it should be treated as a first-class exposure mechanism and analyzed separately.

2 Pattern A: witness-indexed maps (selection by declared public state)

The cleanest receipt-grade pattern is to make selection a deterministic function of an explicit, published witness W . Examples: bucket boundaries by public “epoch class,” coarsening maps by declared overlay configuration, or prefix rules by public staleness class.

Definition 1 (Witness-indexed map family). *Let W be a published witness taking values in $\mathcal{W}$. A witness-indexed family is a collection of deterministic maps $\{f_w\}_{w \in \mathcal{W}}$ (e.g., bucket maps b_w or coarsenings c_w) and a rule $Y = f_W(X)$. Receipts publish (i) the witness value W (or its digest-bound declaration id) and (ii) the chosen map id (digest) $\mathbf{map_id}(W)$.*

Why this is conservative. If W is public and (by declaration) independent of the secret, then publishing W is not avoidable—it is already part of the claim surface. Selection adds no extra secret dependence beyond what is accounted for by conditioning on W . Formally, Lemma ?? in

Synthesis 18 bounds joint leakage of (W, Y) by “leakage of W ” plus worst-case conditional leakage of Y given W .^[?]

Remark 1 (State-conditioned variants in the worked example). *Synthesis 17 includes “state-conditioned” variants (e.g., prefix-fetch equalization conditioned on a bucket-staleness class) that instantiate this pattern end-to-end: the witness is published explicitly, and the receipt claims are conditional.*^[?]

3 Pattern B: pilot/frozen phases (separate selection from audit)

When W is not available (or the map must be chosen from observed behavior), a second conservative pattern is *pilot/frozen* operation: use a pilot window to choose a map, then freeze it and certify it from fresh logs.

Pilot window. Choose a map using internal logs from a declared pilot period. To keep the receipt surface disciplined, publish the resulting map as a digest-bound object and treat it as a *menu change*.

Frozen-audit window. Run the deployment with the frozen map and collect audit logs. Issue certificates for the resulting discrete projection using Synthesis 25. If the map changes again, treat the change as a new audit event and allocate statistical failure probability via Synthesis 26.^[?]

Relation to data splitting / selective inference. Pilot/frozen is a conservative form of *data splitting*: selection is done on one dataset and inference/audit on another. The series does not import post-selection inference machinery, but the warning is shared: publishing a choice made from data can itself leak information and can invalidate naive confidence accounting.^[?]

What this does *not* solve. Publishing a map chosen from pilot logs can leak information about that pilot window. This archive’s conservative stance is: either (i) the pilot window is outside the modeled adversary exposure (explicitly), or (ii) the map-selection output is itself treated as an exposure surface to be bounded (new line item / new interface projection). In either case, the change-control boundary makes the scope reviewable.

4 Receipt-facing checklist for data-dependent parameters

A paper that uses a data-dependent map should be able to cite this note and remain short. Receipts (or ENF+receipt bundles) should include:

- The **map object** (bucket boundaries or partition table) as a digest-bound artifact (`map_id`).
- The **selection discipline**: witness-indexed (publish W / `state_decl_id`) or pilot/frozen (publish pilot scope, freeze date, and event index t).^[?]
- The **audit evidence reference**: Synthesis 25 for the finite-sample certificate, and Synthesis 26 for repeated/expanding audits.^[?, ?]
- An explicit statement of **ENF discipline**: finer telemetry used during pilot is not retained/exposed beyond the declared receipt surface.^[?]

References

- [1] Anonymous. Trace Interfaces for Anonymous DHT Deployments: Declared Projections and ENF Discipline. Synthesis 3, The-One-True-Archive (2026).
- [2] Anonymous. Anonymous DHT Profiling and the Equalization Contract. Anonymity B, The-One-True-Archive (2026).
- [3] Anonymous. Discrete Equalization Audits from Logs: Menu-Level Finite-Sample TV Certificates. Synthesis 25, The-One-True-Archive (2026).
- [4] Anonymous. Anytime Statistical Spending for Menu Audits: Change-Control-Safe Confidence Budgets. Synthesis 26, The-One-True-Archive (2026).
- [5] Anonymous. Bucket Design for Continuous Timing Interfaces: Making “Continuous” Telemetry Auditably Discrete. Synthesis 27, The-One-True-Archive (2026).
- [6] Anonymous. Coarsening and Large-Alphabet Projections for Log Audits: Safe Partitioning Rules that Keep Certificates Sample-Feasible. Synthesis 28, The-One-True-Archive (2026).
- [7] Anonymous. Plan and State Equivalence: Digest-Bound Surfaces and Conditional Accounting. Synthesis 18, The-One-True-Archive (2026).
- [8] Anonymous. Worked Example: Receipt/Artifact Interlock for OHTTP + DHT Profiling (with optional variants). Synthesis 17, The-One-True-Archive (2026).
- [9] J. D. Lee, D. L. Sun, Y. Sun, and J. E. Taylor. Exact post-selection inference, with application to the lasso. *The Annals of Statistics*, 44(3):907–927, 2016. DOI: 10.1214/15-AOS1371. <https://projecteuclid.org/journals/annals-of-statistics/volume-44/issue-3/Exact-post-selection-inference-with-application-to-the-lasso/10.1214/15-AOS1371.full>.